* Frame buffer : Contain all pixel intensity values

See 2 * no of bits/pixel = depth of buffer = no of bits/plane

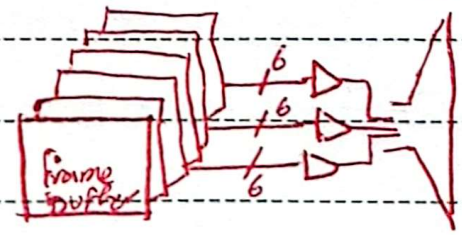
* pixel Color Representation (Color Palette)

* Color Palette : the set of Colors that display sys Can use to represent an image

* Two Common methods are used :

[1] Direct Coding : each pixel store its own Color value directly

* if frame buffer support 6 planes of memory (2 for each Color) then 2^2 level of R, G & B



* for true Color image, Color depth 24 bit means 24 planes of memory (8 for each Color) then 2^8 level of R, G, B, Total $2^{24} \simeq 16$ M Color pallette

Ex. if we use direct Coding RGB with 3 bits per primary Color how many possible Colors do we have for each pixel

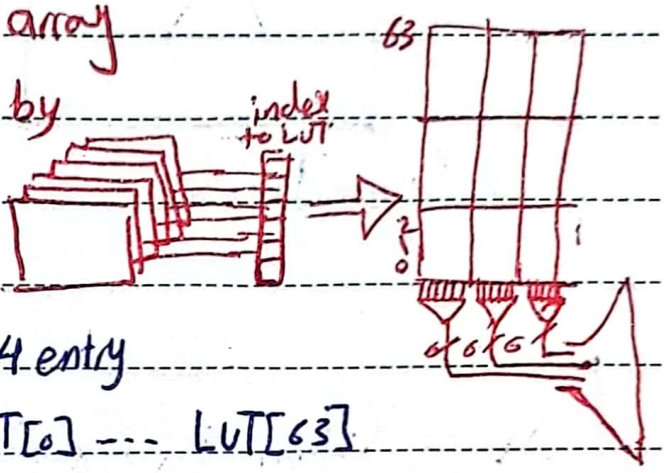
Sol

no of Colors for each pixel = $2^3 \times 2^3 \times 2^3 = 2^9 = 512$ Color

(2) indexed Color & Look up table (LUT)

instead of storing the full color value in each pixel, the pixel hold an index that points to a Color in LUT.

* reduce memory & support large no of Colors
* each elements in the array represent a Color, indexed by position within array



* if pixel depth = 6

the index of table is $= 2^6 = 64$ entry (no of entries) Say LUT[0] --- LUT[63]

EX.2 if we use 10 bits/pixel value in 24 bit LUT how many entries does LUT have & how much size of it in bytes

Sol no of entries $= 2^{\text{no of bit/pixel}} = 2^{10} = 1024$ entry

LUT size $= (\text{no of entries} * \text{no of bits in LUT}) / 8 = (2^{16} * 24)/8$

$= 3072$ byte

EX.3 if we use 2 bytes pixel value in a 24 bit LUT representation, how many entries & size of LUT.

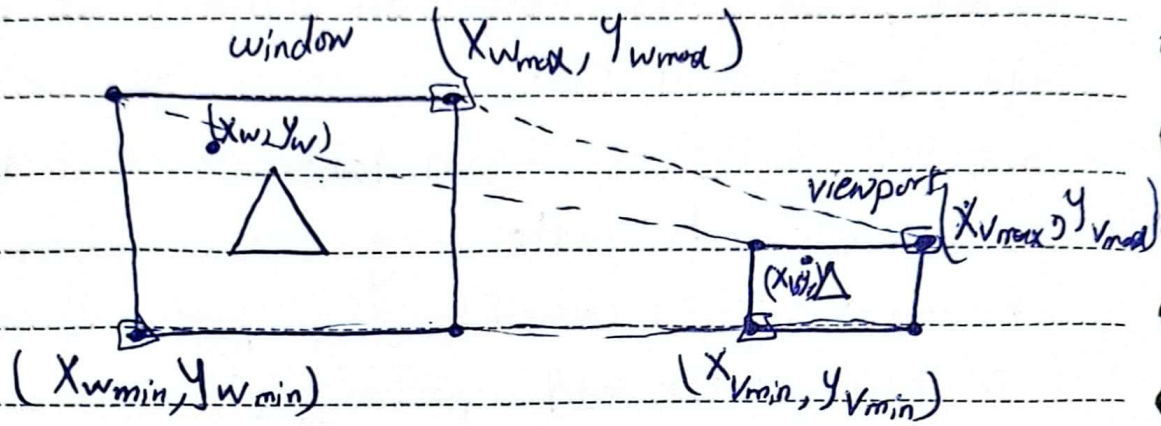
Sol 2 bytes $= 2*8 = 16$ bit

LUT entries $= 2^{16} = 65536$ entry

LUT size $= (2^{16} * 24)/8 = 196\,608$ byte

Sec 3

* window to viewport mapping



$$X_v = X_{vmin} + (S_x * (X_w - X_{win}))$$

$$Y_v = Y_{vmin} + (S_y * (y_w - y_{wmin}))$$

where Sx, Sy are scaling factor

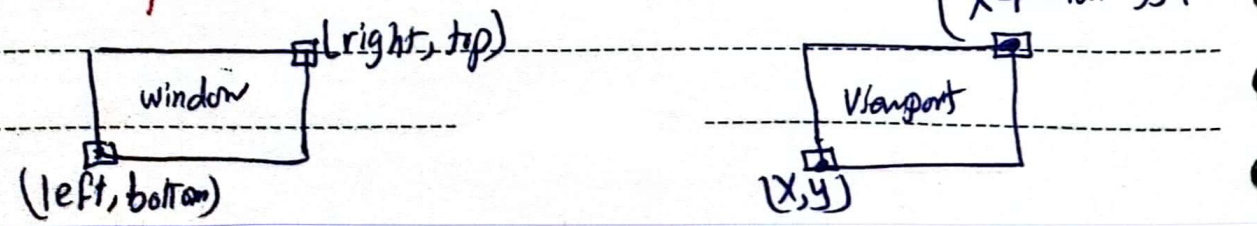
$$S_x = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}} \qquad S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{vmin}}$$

Set window :

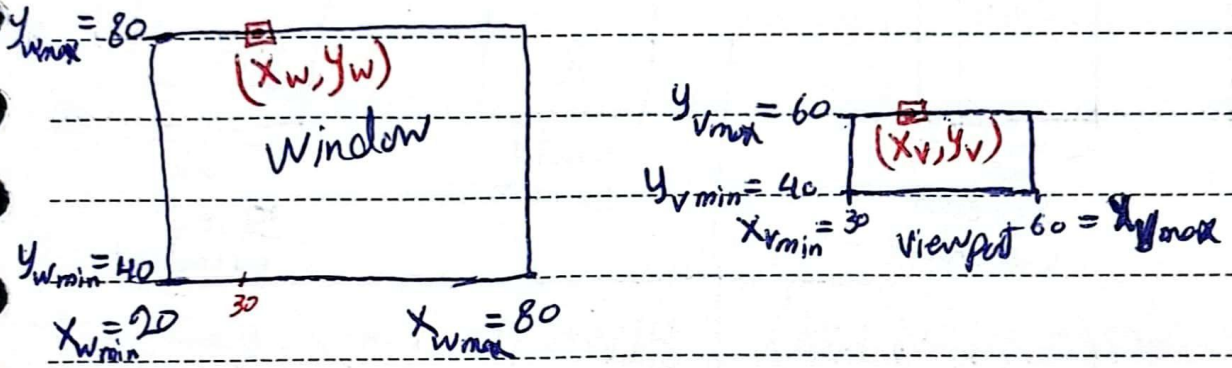
glOrtho2D(left, right, bottom, top)

set viewport

glViewport (x, y, width, height)



Ex.1 Consider the window & viewport, the window has $(W_L, W_r, W_b, W_t) = (20, 80, 40, 80)$ & viewport $(V_L, V_r, V_b, V_t) = (30, 60, 40, 60)$ for $(X_w, y_w) = (30, 80)$ find (X_v, Y_v)



Sol

$$S_x = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}} = \frac{60 - 30}{80 - 20} = 0.5$$

$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}} = \frac{60 - 40}{80 - 40} = 0.5$$

$$X_v = X_{vmin} + (S_x (X_w - X_{wmin})) = 30 + 0.5(30 - 20)$$
$$= 35$$

$$y_v = y_{vmin} + (S_y (y_w - y_{wmin})) = 40 + 0.5(80 - 40)$$
$$= 60$$

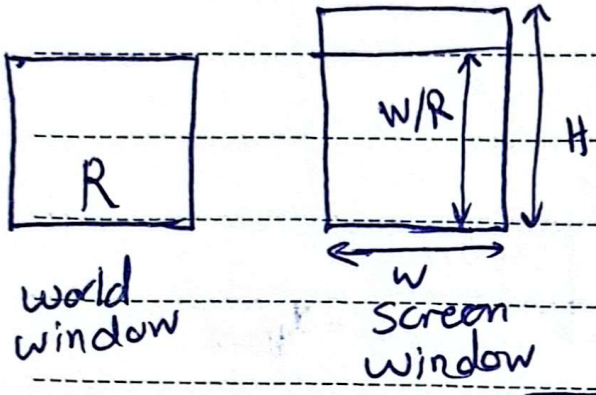
```
glOrtho2D(20, 80, 40, 80)   //Set window
glViewport(30, 40, 30, 20)  //Set viewport
```

x, y, (width) (60-30), (height) (60-40)

* setting the window & viewport Automatically:-

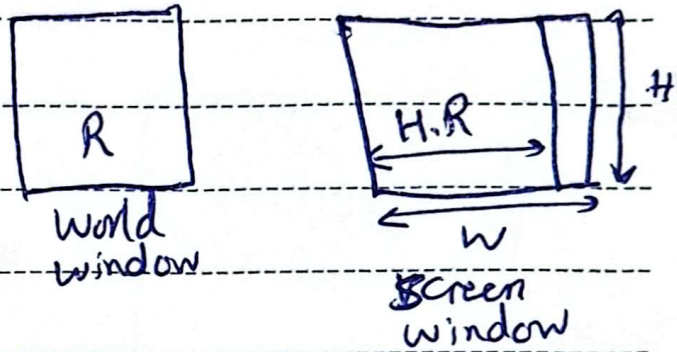
(R) aspect Ratio of world, (W, H) is width & height of screen window

if $R > W/H$



glViewport(0, 0, W, W/R)

if $R < W/H$



glViewport(0, 0, H*R, H)

Ex.1 Find the matching viewport for a world window with aspect Ratio R = 1.6 when the screen window has width 360 & height 200

Sol $R = 1.6 < \frac{W_v}{H_v} = \frac{360}{200} = 1.8$

then $H_v = H = 200$ & $W_v = H * R = 200 * 1.6 = 320$

glViewport(0, 0, 320, 200)

Ex.2 if $R=2$ & $W_v=360$, $H_v=200$

Sol $R=2 \Rightarrow \frac{W_v}{H_v}=\frac{360}{200}=1.8$

then $W_v=W=360$ & $H_v=W/R=\frac{360}{2}=180$

glViewport(0,0,360,180)

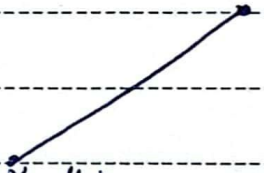
Sec 4 : line drawing

* Cases doesn't need line drawing algorithms

horizontal line	vertical line	$y = \pm x$
$m = 0$	$m = \infty$	$m = \pm 1$

* Aliasing is the jagged edges on curves & diagonal lines in a bitmap image

* Anti-aliasing : is the process of smoothing out those jaggies
  - → graphics sw prog have options for anti-aliasing text & graphics
  - → Enlarging a bitmap image accentuates (highlight) the effect of aliasing



1 DDA Algorithm

the digital differential analyzer is a scan Conversion line algorithm based on using dx or dy

Step 1 : Calculate ΔX, Δy, $M = \Delta y / \Delta x$

Step 2 : Find no of steps = the greates of ($\Delta X, \Delta y$) absolute

Step 3 : ($\Delta x, \Delta y$) لو سالب سالب ($\Delta x, \Delta y$) لو موجب موجب ($\Delta x, \Delta y > 0$)

	Right to left	left to right
$M \leq 1$	$X_{k+1} = \text{round off}(X_k - 1)$ $y_{k+1} = \text{round off}(y_k - m)$	$X_{k+1} = \text{round off}(X_k + 1)$ $y_{k+1} = \text{round off}(y_k + M)$
$M \geq 1$	$X_{k+1} = \text{roundoff}(X_k - \frac{1}{m})$ $y_{k+1} = \text{roundoff}(y_k - 1)$	$X_{k+1} = \text{roundoff}(X_k + \frac{1}{m})$ $y_{k+1} = \text{roundoff}(y_k + 1)$
$M = 1$	$X_{k+1} = \text{roundoff}(X_k - 1)$ $y_{k+1} = \text{"} \quad (y_k - 1)$	$X_{k+1} = \text{roundoff}(X_k + 1)$ $y_{k+1} = \text{"} \quad (y_k + 1)$

Ex.1 Calculate the points bet. $(5,6)$ and $(8,12)$ — start point (x_1, y_1), end point (x_2, y_2)

Sol

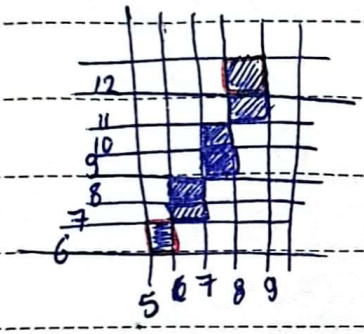
① $\Delta X = 8-5 = 3$, $\Delta Y = 12-6 = 6$, $M = \frac{\Delta Y}{\Delta X} = \frac{6}{3} = 2$

② no of steps $= \Delta y = 6$

③ $M > 1$ & drawn from left to right $= \Delta Y, \Delta X > 0$

then increase $y+1$, Calculate $X = \text{round off}(X + \frac{1}{m})$ → $\frac{1}{2}$

X_k	y_k	X_{k+1}	y_{k+1}	round off (x,y)
5	6	5.5	7	(6,7)
		6	8	(6,8)
		6.5	9	(7,9)
		7	10	(7,10)
		7.5	11	(8;11)
		8	12	(8,12)



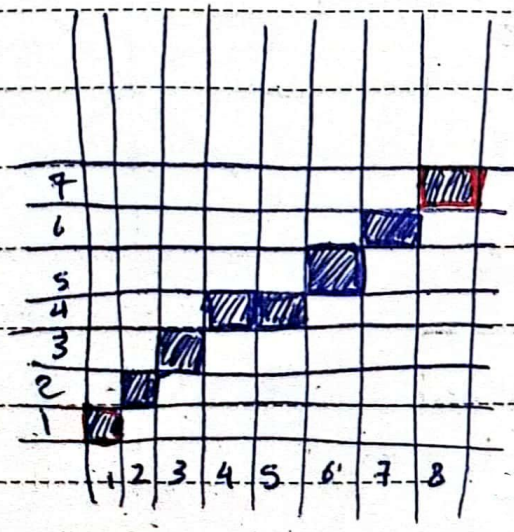
Ex.2 draw line from $(1,1)$ to $(8,7)$ using DDA algorithm — x_1, y_1 to x_2, y_2

① $\Delta X = 8-1 = 7$, $\Delta Y = 7-1 = 6 > 0$ left to Right

② $M = \frac{6}{7} = 0.857 < 1$, no of steps $= 7$

③ increase X, Calculate $y = \text{rdff}(y+m)$

X_{k+1}	y_{k+1}	round off (x,y)
2	1.857	(2,2)
3	2.717	(3,3)
4	3.57	(4,4)
5	4.43	(5,4)
6	5.28	(6,5)
7	6.14	(7,6)
8	7	(8,7)



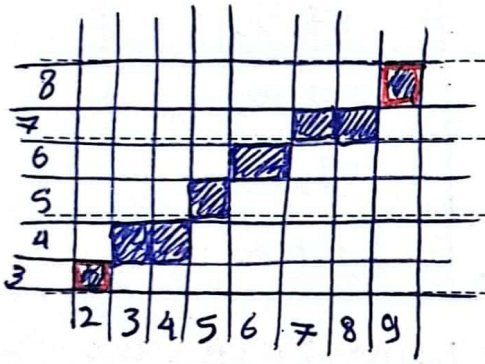
Ex.3 use DDA algorithm to draw a line from $(x_1, y_1) = (9,8)$ to $(x_2, y_2) = (2,3)$

Sol

① $\Delta X = 2-9 = -7$, $\Delta Y = 3-8 = -5 < 0$ from right to left

② $m = \frac{-5}{-7} = 0.714 < 1$ no of steps = 7

③ Increas X, $y = y_k - m$



x_{k+1}	y_{k+1}	round off (x_{k+1}, y_{k+1})
8	7.28	(8,7)
7	657	(7,7)
6	5.8	(6,6)
5	5.14	(5,5)
4	4.4	(4,4)
3	3.7	(3,4)
2	3	(2,3)

---

2. Bresenham – line drawing algorithm.

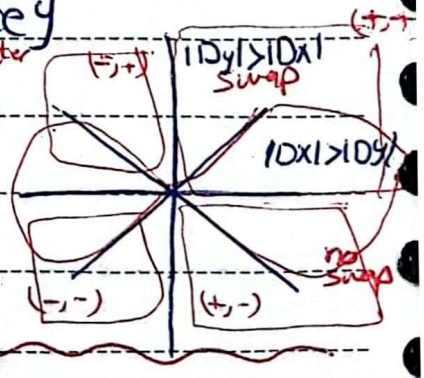
this method is described for a line segment with $m = 0:1$

So we work by inc or dec. X & Calculate y

step 1: Calculate ΔX, ΔY, $P_0 = 2\Delta Y - \Delta X$ ← decision Parameter

step 2: if

①	$P_k < 0$	$P_{k+1} = P_k + 2\Delta Y$ inc(X), $y = y_{k+1}$
②	$P_k >= 0$	$P_{k+1} = P_k + 2\Delta Y - 2\Delta X$ inc(x), inc(y)



step 3: repeat until endpoint is reached or no of iterations = $(\Delta X - 1)$ times

if $|\Delta y| > |\Delta x|$ swap
if $\Delta X < 0$ dec
$\Delta y < 0$ dec

$2\Delta y = 16$ ، $2\Delta y - 2\Delta x = -4$

EX.1 start point $(x_1, y_1) = (20, 10)$ endpoint $(x_2, y_2) = (30, 18)$ ①

Sol $\Delta x = 30 - 20 = 10$, $\Delta y = 18 - 10 = 8$

$p_0 = 2\Delta y - \Delta x = 16 - 10 = 6 >= 0$? inc x, inc y

	P_{k+1}	X	y	setpixel (X,y)
	-	20	10	(20,10)
Case 2	6 →	21	11	(21, 11)
Case 2	2 →	22	12	(22,12)
Case 1	-2 →	23	12	(23, 12)
Case 2	14 →	24	13	(24, 13)
Case 2	10 →	25	14	(25 ,14)
Case 2	6 →	26	15	(26, 15)
Case 2	2 →	27	16	(27, 16)
Case 1	-2 →	28	16	(28 , 16)
Case 2	14 →	29	17	(29 ,17)
Case 2	10 →	30	18	(30,18)

كل ما x زادت 1 زيد



Ex.2 1'st point $(x_1, y_1) = (3, 32)$ & 2nd point $(x_2, y_2) = (8, 38)$

① $\Delta X = 8-3 = 5$, $\Delta y = 38-32 = 6$

$\Delta y > \Delta X$ ⟶ (swap x, y)

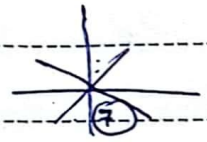
$(32, 3)$ & $(38, 8)$ $\Delta X = 6$, $\Delta y = 5 > 0$ (in I only)

then $P_0 = 2\Delta y - \Delta X = 10-6 = 4$ Case 2 ⑦ $P_{k+1} = P_k + 2$ ($2\Delta y - 2\Delta X$)

Case 1 ⑥ $P_{k+1} = P_k + 10$ ($2\Delta y$)

	P_{k+1}	X	y	Point (y, x)
	–	32	3	(3, 32)
Case 2	4 →	33	4	(4, 33)
〃	2 →	34	5	(5, 34)
〃	0 →	35	6	(6, 35)
Case 1	−2 →	36	6	(6, 36)
Case 2	8 →	37	7	(7, 37)
〃	6 →	38	8	(8, 38)

Ex.3 $(8, 32)$ & $(3, 38)$



① $\Delta X = 3-8 = -5$, $\Delta y = 38-32 = 6$

$|\Delta y| > |\Delta X|$ ⟶ swap (x, y)

$(32, 8)$ & $(38, 3)$ ⟶ ΔX +ve, $\Delta y = |-5|$ ⟶ dec y

$P_0 = 2\Delta y - \Delta X = 10-6 = 4$

P_{k+1} ① ≥0 ② <0

$P_k - 2$ $P_k + 10$

	P_{k+1}	X	y	set (y, x)
	–	32	8	(8, 32)
Case ①	4	33	7	(7, 33)
Case ②	2	34	6	(6, 34)
〃	0	35	5	(5, 35)
Case 2	−2	36	5	(5, 36)
	8	37	4	(4, 37)
	6	38	3	(3, 38)

CamScanner

Ex.4 $(x_1, y_1) = (38, 3)$ & $(x_2, y_2) = (32, 8)$

① $|\Delta x| = |32 - 38| = |-6| = 6$, $|\Delta y| = 8 - 3 = 5$

$\therefore \Delta x > \Delta y$ no swap

$\therefore \Delta x = -6$ x is dec. , y inc.

$P_0 = 2\Delta y - \Delta x = 10 - 6 = 4$

≥ 0 P_{k+1} < 0

$P_k - 2$ $P_k + 10$

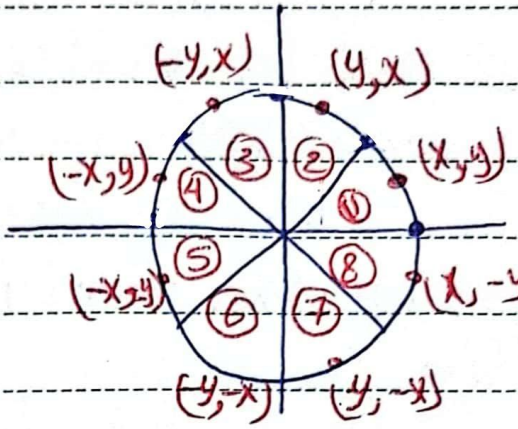
P_{k+1}	x	y	set (x, y)
-	38	3	(38, 3)
4	37	4	(37, 4)
2	36	5	(36, 5)
0	35	6	(35, 6)
-2	34	6	(34, 6)
8	33	7	(33, 7)
6	32	8	(32, 8)

CamScanner

Secs: Circle drawing

* Midpoint algorithm

Caculate points in one octant then drive the rest for the other 7 octants



Step 1: initial point $(0, r)$ (r ↓ radius)

Step 2: $P_0 = 1 - r$

Step 3:

if $P_k < 0$	$y_{k+1} = y_k$ (inc x) $P_{k+1} = P_k + 2x_{k+1} + 1$
$P_k > 0$	$y_{k+1} = y_k - 1$ (inc x) $P_{k+1} = P_k + 2x_{k+1} + 1 - 2y_{k+1}$

Step 4: repeat until $x \geq y$

EX: using Bresenham circle with center (0,0) & $r = 10$, second octant

Sol

① initial point $(0, 10)$

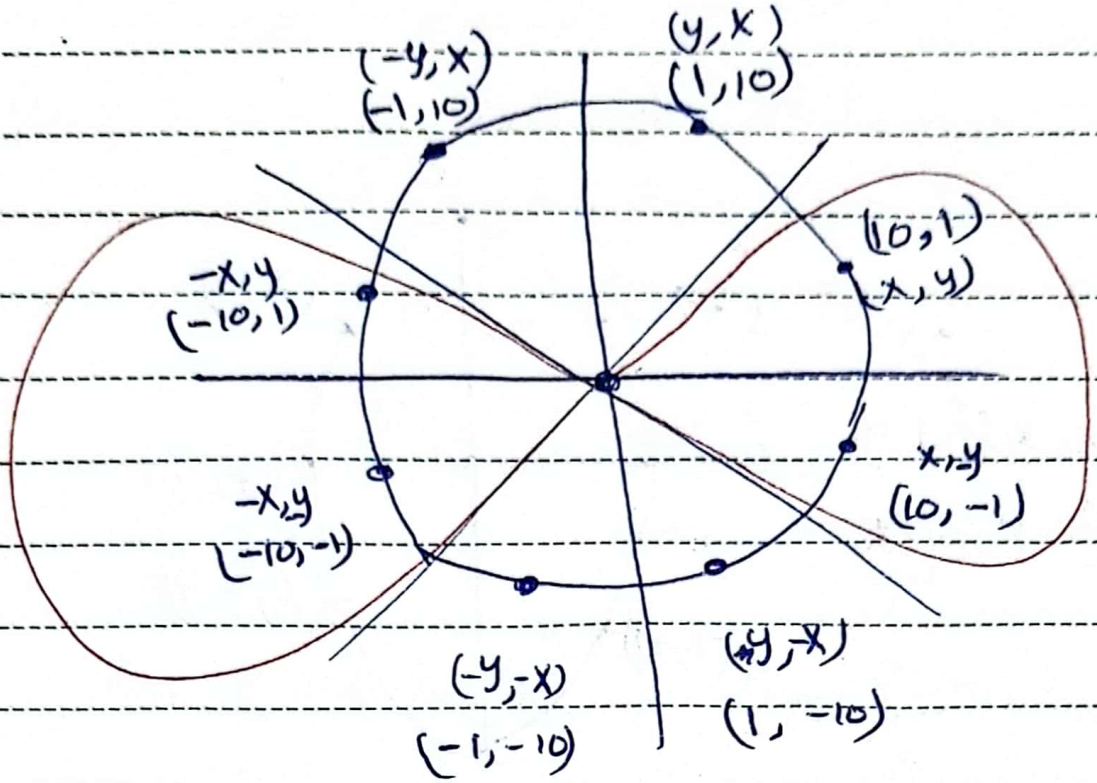
$P_0 = 1 - 10 = -9$

P_k ≥ 0 → $y = y - 1$; $P_{k+1} = P_k + 2x + 1 - 2y$

P_k < 0 → $y = y$; $P_{k+1} = P_k + 2x + 1$

k	P_k	x_{k+1}, y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
0	-9	(1, 10)	2	20
1	$-9+3 = -6$	(2, 10)	4	20
2	-1	(3, 10)	6	20
	$-1+7 = 6$	(4, 9)	8	18
	-3	(5, 9)	10	18
	$-3+11 = 8$	(6, 8)	12	16
	5	(7, 7)	—	—

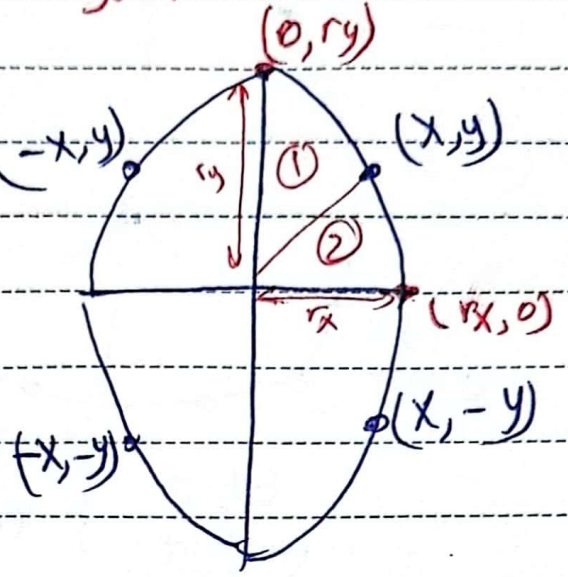
$x \geq y$



if the center wasn't $(0,0)$

then $(x_c + x, y_c + y)$

[2] Ellipse generation algorithm



Calculate

Region 1 (inc x), (Cal. y)

Region 2 (dec y), (Cal x)

Ex 1 Given input ellipse Prometer $r_x = 8, r_y = 6$

Sol

for region 1 ① $(x_0, y_0) = (0, 6)$

② $P1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2 = -332$

P_k

>0 ① <0 ②

$x = x+1$

$y = y - 1$

$P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$

$x = x+1$

$y = y$

$P_{k+1} = P_k + 2r_y^2 x_{k+1} + r_y^2$

$2 \times 6^2 \times 1$

repeat until $2r_y^2 x \geq 2r_x^2 y$ Case 2

k	$P1_k$	(x_{k+1}, y_{k+1})	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
0	-332	(1, 6)	72	768
1	-224	(2, 6)	144	768
2	-44	(3, 6)	216	768
3	208	(4, 5)	288	640
4	-108	(5, 5)	360	640
5	288	(6, 4)	432	512
6	244	(7, 3)	504 >	384

STOP

R.2 النقطة اللي انتقلت عليها من

for Region 2 $(x_0, y_0) = (7, 3)$

$$P_0^2 = r_y^2 (x_0 + \tfrac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$$

P_k

> 0 ← → < 0

$x = x$
$y = y - 1$

$P_{k+1} = P_k - 2r_x^2 y_{k+1} + r_x^2$

$x = x + 1$
$y = y - 1$

$P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$

repeat until we reach point $(0, r_x)$

K	P_K^2	(x_{k+1}, y_{k+1})	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
0	-23	(8,2)	576	256
1	361	(8,1)	576	128
2	297	(8,0)	-	-

Stop